



Learning Objectives

In this tutorial you will explore some modelling towards optimization problems and using the techniques we have learned to analyze the results.

These problems relate to the following course learning objectives:

- *Model optimization problems.*
- *Recognize what type of optimization problem they need.*
- *Use Jupyter Notebook to approximate the solution.*
- *Create a visualization of the problem and the solution.*

Problems

1. You recently started playing board games and your board game collection is getting large enough that you decided to buy a bookshelf to store your games.

You don't have a lot of money, so you find a cheap used 2×2 Ikea Kallax bookshelf. The dimensions of each of the four shelves is $35\text{cm} \times 35\text{cm} \times 35\text{cm}$.

Your board game collection consists of 19 games and their boxes have a square profile $30\text{cm} \times 30\text{cm}$ with the following thicknesses:

- 5 games with box thickness of 7cm
- 10 games with box thickness of 8cm
- 4 games with box thickness of 9cm

- (a) What is the optimal way to place them in your new bookshelf?
- (b) Report the solution.
- (c) Does the solution change if one of the shelves came with a defect and was 36cm wide instead?

2. You live in a condo building and want to **optimize** the resting position of the elevators.

Here are some details about the building:

- There are 3 elevators
- The building has 30 floors with the ground floor on floor 0 and apartments on floors 1–30 (inclusive).
- There are 10 apartments per floor with 2 people living in each apartment.
- The elevators take 1 second to open/close the doors

- (a) Use the DABAR framework to solve this problem.
- (b) Create a visualization of the problem and solution.
- (c) Calculate the sensitivity with respect to the number of people in each floor and with respect to the time it takes the elevators to open/close the doors